- The integration layer: The HL7 engine

Deployment and use in various fields

OpenVista

OpenVista has been deployed at about 1500 care sites. OpenVista (R) source code is freely available, and it could be as effective and low-cost an option for even non-Veterans Affairs hospitals that are seeking to implement a complete hospital information system, but have been held back due to cost. OpenVista is the largest medical record system in the US, serving nearly 700 hospitals and nursing homes.

OpenVista (R) can be customised for large hospitals as well as small medical practices, and can be used across the continuum of acute, ambulatory and long-term care environments. It is suitable for use in multi-facility, multi-specialty healthcare organisations. NewCreationIT can customise, install, run and support OpenVista (R) for small clinics/medical practices or large hospitals, and can include Web and mobile add-ons. OpenVista (R) is a registered trademark of Medsphere Systems Corporation. NewCreationIT deploys, customises and supports the open source OpenVista software for medical practices and hospitals, globally. NewCreationIT is not affiliated with, or approved by, Medsphere Systems Corporation.

Open MRS

Countries like Kenya, Rwanda, Tanzania, South Africa, Lesotho, Zimbabwe, Uganda and Pakistan have the most number of Open MRS deployments. It is used for HIV treatment in Kenya, Rwanda and Tanzania. Each site has requirements that are different from the base system. It is a Herculean task for the designers to satisfy the diverse requirements, so they built the system with extensibility in mind. Thus, whenever deploying Open MRS to record data for different diseases, a new version is not released—instead, implementers use the extensible data model and modules to build a custom MRS system atop the base system.

Open EMR

Open EMR has been extensively used in the US. It is notably the best EHR for small healthcare units. However, practitioners from countries like Kenya, Puerto Rico, Australia, India, Pakistan, Bermuda and Nepal are getting savvy and adopting this system.

Challenges faced while using these software

OpenVista

The main challenge is the decentralised structure of the OpenVista system. VistA files are maintained locally at the VA medical facilities where they are generated. Permission for access must be granted by each facility included in an analysis. Access to identifiable patient-level data must also be approved by each local Institutional Review Board (IRB). For multi-site studies, gaining access can be a complex, time-consuming process. The second challenge is whether to keep the current MUMPS database, or migrate to a relational or object oriented database. Migration presents additional challenges. Local VistA implementations store data in MUMPS volume sets on pre-allocated disk sections. When volume sets exceed 16 GB, performance degrades, due to accelerated data accumulation, and growing number of physicians demanding availability of all records. Nevertheless, not all data elements necessary to an investigation may be available from the same file. Besides, some may exist only as text, not in a convenient, computable format. Therefore, the complexity of, and the site-to-site variations in OpenVista files make local technical assistance a necessity. One further complication is the variability of bandwidth and network reliability.

Open MRS

This health record system can not only handle thousands of patients and hundreds of thousands of observations, but is also scalable to tens of thousands of patients and millions of observations. This uses a lot of *Update* and *Read* operations. Tomcat and the JVM allocate memory to a Web app each time one uses *Update* or *Read* functions in the Web Application Manager. When the app is destroyed or recreated, some of this memory may not be released or lead to memory leakage. If you update or read the Web app too many times, Tomcat may run out of allocated memory, and will stop responding. Note the following error in the Tomcat logs:

```
java.lang.OutOfMemoryError: PermGen space
```

You can sort this out by allowing Tomcat to use more memory, or by restarting the Tomcat server at intervals, if you have to repeatedly update or read from the database.

Open EMR

Installation can be problematic if the instructions are not carefully followed. The system works on all known distributions of Linux, Mac Os X, Windows 2003 server and Windows XP Professional. The installation on Windows 2000, Windows XP Home Edition, FreeBSD and OpenBSD are less friendly and usually take quite a bit of troubleshooting. Its somehow less user friendly, thats the challenge users generally face using it. END

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